

VASU SHARMA

Graduate Structural Engineer – Politecnico di Milano

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PROFILE

Graduate Structural Engineer (MSc, Politecnico di Milano) with a strong foundation in structural analysis and design of RC and steel structures, including seismic design (Eurocode/NTC). Experienced in FEM-based modelling and analysis of building and bridge systems (SAP2000, ETABS, MIDAS Gen), with additional expertise in nonlinear analysis using OpenSees. BIM-capable (Revit/AutoCAD) with basic exposure to automation.

KEY SKILLS

Nonlinear structural analysis • Seismic design (EU Codes/EC8/NTC-based work) • RC/steel modelling
FEM/FEA workflows • Bridge design fundamentals (PT decks, moving loads) • BIM delivery • Technical reporting

TECHNICAL TOOLS

Structural analysis: OpenSees | CSI SAP2000 | CSI ETABS | MIDAS GEN | CSI SAFE | STAAD.Pro | Tekla
Computation/automation: MATLAB | Python (basic, upskilling) | Dynamo for Revit (learning/exposure)
BIM & drafting: Autodesk Revit | Autodesk AutoCAD

EXPERIENCE

Structural Engineer – Vasundhara Infrastructure and Construction LLP | Nov 2025 - Present

- Currently working on design and analysis of a G+9 storied building design.
- Superstructure model in **ETABS**; Analysis in **SAP2000/MIDAS GEN**; foundation design in **SAFE**.
- BIM-based structural detailing in **Revit**, reinforcement detailing and bar bending schedules (BBS)

Industrial Training Research Intern - CSIR-CBRI, Roorkee (India) | Jul 2021 – Jan 2022

- Supported concept development for rapid housing / post-disaster construction and tensegrity-based systems
 - Assisted model fabrication and preparation for dynamic testing to support design iterations
 - Prepared technical documentation and presentation material for design updates
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EDUCATION

MSc Structural Engineering - Politecnico di Milano, Italy | Oct 2022 – 2025 | Final grade: 100

Erasmus Exchange - IIT Bombay (Indian Institute of Technology Bombay), India | Jul 2023 – Jan 2024

BE Civil Engineering - Thapar Institute of Engineering & Technology, Patiala (India) | 2018 – 2022 | CGPA: 9.02/10

PROJECTS

MSc Thesis - Nonlinear modelling of corroded RC columns under cyclic loading (Polimi) | 2024 – 2025

- Built numerical models and ran parametric studies in **OpenSees (TCL/OpenSeesPy)**
- Interpreted hysteretic trends (stiffness/strength degradation) and documented results clearly

Bridge Design - 40 m post-tensioned box girder deck **CSI SAP2000** | Aug 2024 – Jan 2025

- Developed analysis model and assessed behaviour under key actions/load combinations
- Studied response under moving loads; produced governing outputs for checks and documentation

Earthquake Resistant Design Projects (MIDAS / SAP2000/ ETABS) | Aug 2023 – May 2024

- Designed a 4-storey dual-system RC building following Eurocode 8 and NTC-18 principles
- Evaluated seismic vulnerability of an existing masonry building and summarised findings

Bachelor's Thesis - G+10 composite building (**STAAD.Pro** + **ETABS** + **Revit**) | Jun 2021 – Jun 2022

- Code-based modelling and analysis of a multi-storied building using FEM (**ETABS**)
 - Developed BIM model/drawings in Revit to support coordination and design communication
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LEADERSHIP & ACTIVITIES

Student Placement Representative (CILP), Thapar Institute | Representative: Student Consultative Committee

Hobby: Piano | Athletics: Gold medals (100m/200m/400m/Long Jump)

LANGUAGES

English (Native) | Italian (A2 – Upskilling) | Spanish (A2)